

Problem formulation and optimality conditions

Jen's total dose must satisfy $a_1x_1 + a_2x_2 = 1$ with $a_1 = 1$, a_2 as given. We minimise $T = x_1 + x_2$ subject to $x_1, x_2 \geq 0$ and $x_1 + a_2x_2 = 1$. This is a simple linear program. If $a_2 > 1$, the second therapy is faster (higher rate), so the optimum uses only therapy 2: $x_1^* = 0$ and $x_2^* = 1/a_2$. If $a_2 < 1$, the optimum uses only therapy 1: $x_1^* = 1$, $x_2^* = 0$. (If $a_2 = 1$, any split with $x_1 + x_2 = 1$ is optimal.) In our case the code shows $a_2 > 1$ (see below), so the unique minimiser is $(x_1^*, x_2^*) = (0, 1/a_2)$.

Computing a_2 from the code

The code defines $a_2 = 2f + \frac{1}{2}$, where

$$f = 72^{-10^{24}} + \sum_{k=3}^{n+2} \left(\frac{1}{2}\right)^k - \sum_{k=3}^{n+2} 72^{-10^{24}k}.$$

As $n \rightarrow \infty$, $\sum_{k=3}^{\infty} (1/2)^k$ is an infinite geometric series with first term $a = (1/2)^3$ and ratio $r = 1/2$. Since $|r| < 1$, its sum is $a/(1-r) = (1/8)/(1/2) = 1/4$ ¹. The subtracted terms $72^{-10^{24}k}$ are astronomically small, and the initial $72^{-10^{24}}$ is likewise essentially 0. Thus

$$f \approx \frac{1}{4} + (\text{tiny positive}) - (\text{even tinier}) > \frac{1}{4},$$

so $2f + 1/2 \approx 1 + (\text{tiny positive})$. In particular $a_2 > 1$.

Optimal solution and distance calculation

Because $a_2 > 1$, the optimal schedule uses only the faster therapy: $x_1^* = 0$, $x_2^* = 1/a_2$ (so $a_1x_1^* + a_2x_2^* = 1$). Numerically $1/a_2 \approx 1$ (up to an infinitesimal). The provided solution $(0.1, 0.9)$ differs from $(0, 1/a_2)$ by

$$|0.1 - 0| = 0.1, \quad |0.9 - 1/a_2| \approx |0.9 - 1| = 0.1.$$

The infinity norm of the difference is $\max(0.1, 0.1) = 0.1$. This is less than 0.15.

Answer: Yes. The optimum is effectively $(0, 1)$ (since $a_2 \approx 1^+$), and $(0.1, 0.9)$ is at distance 0.1 in sup-norm, which is ≤ 0.15 .

¹ Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series